ECCB 2026 · TUTORIAL · SESSION 3

Building Agentic LLM Workflows for Biomedical Knowledge Graphs

Tools, MCP, and reusable skills for querying structured biological data

Biomedical Informatics Group

How to develop and query custom knowledge graphs using LLMs and multi-omics

THE WORKFLOW SO FAR

What you’ve built so far

Idea
“Which factor separates the subtypes?”

→

Hard-coded code
You write the analysis

→

Interpretation
You read the factors, weights, and R^2

- ✓ Deterministic and repeatable
- ✓ Every step is a line you wrote

- ✗ You must know the schema first
- ✗ New question → new code

THE ALTERNATIVE

Alternative - LLM agents

Idea
"Which factor separates the subtypes?"

→

Hard-coded code
You write the analysis

→

Interpretation
You read the factors, weights, and R^2

- ✓ Deterministic and repeatable
- ✓ Every step is a line you wrote
- ✗ You must know the schema first
- ✗ New question → new code

↓

Natural-language question

→

LLM agent
THE AGENT writes it

→

Interpretation
THE AGENT reads it — you review

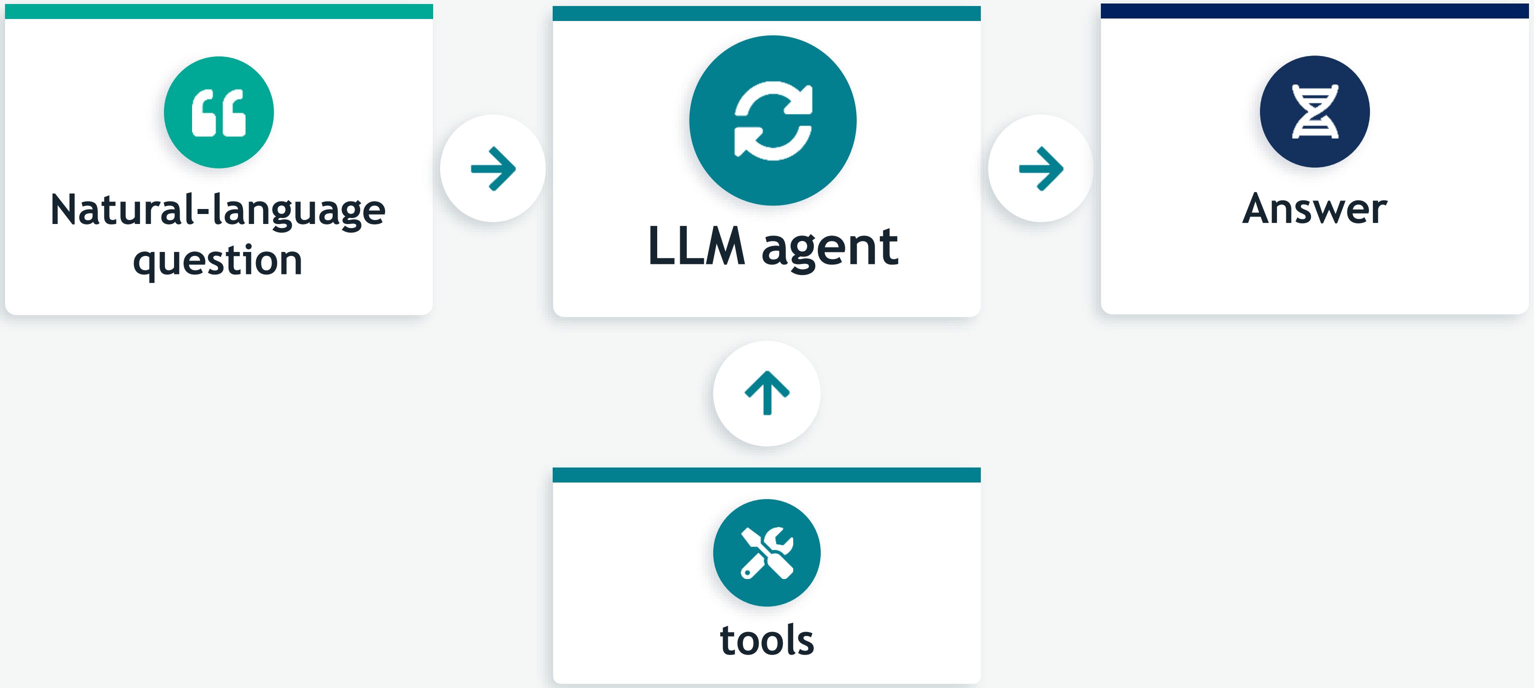
- ✓ No translation step — ask in English
- ✓ Route chosen after seeing results
- ✗ Non-deterministic between runs
- ✗ You must still read the trace

CORE CONCEPT

What is an agent?

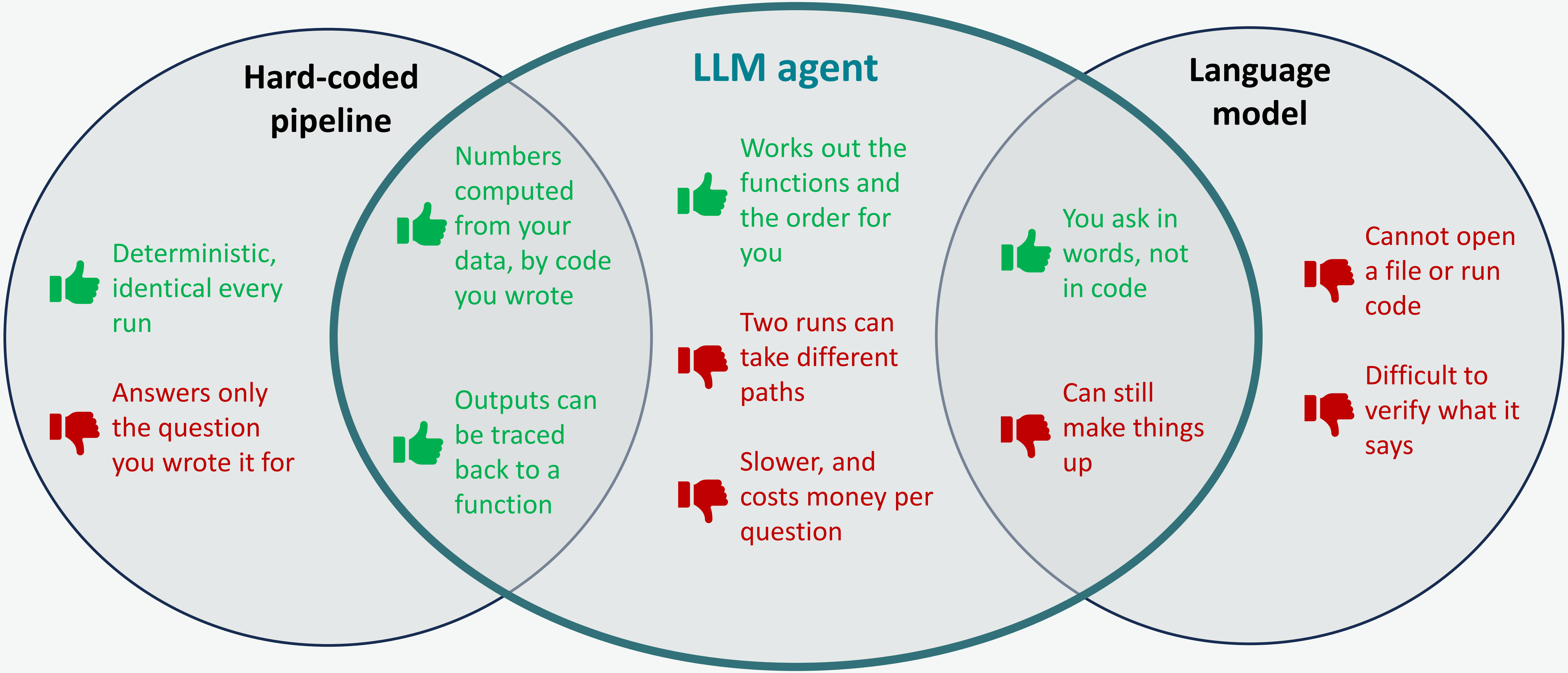
"An agent is a **LLM** that **runs tools in a loop** to achieve a goal"

- Simon Willison



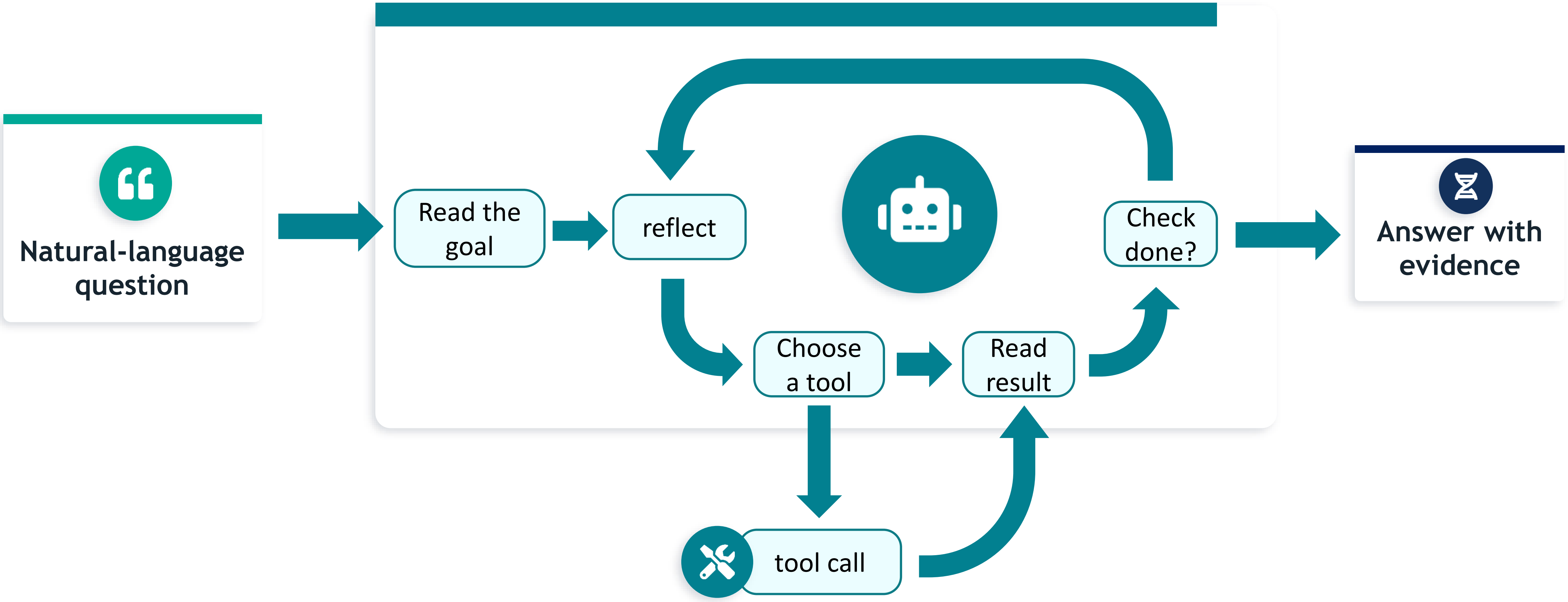
WHY AN AGENT

Where an agent sits



THE AGENT LOOP

How does the agent work under the hood?



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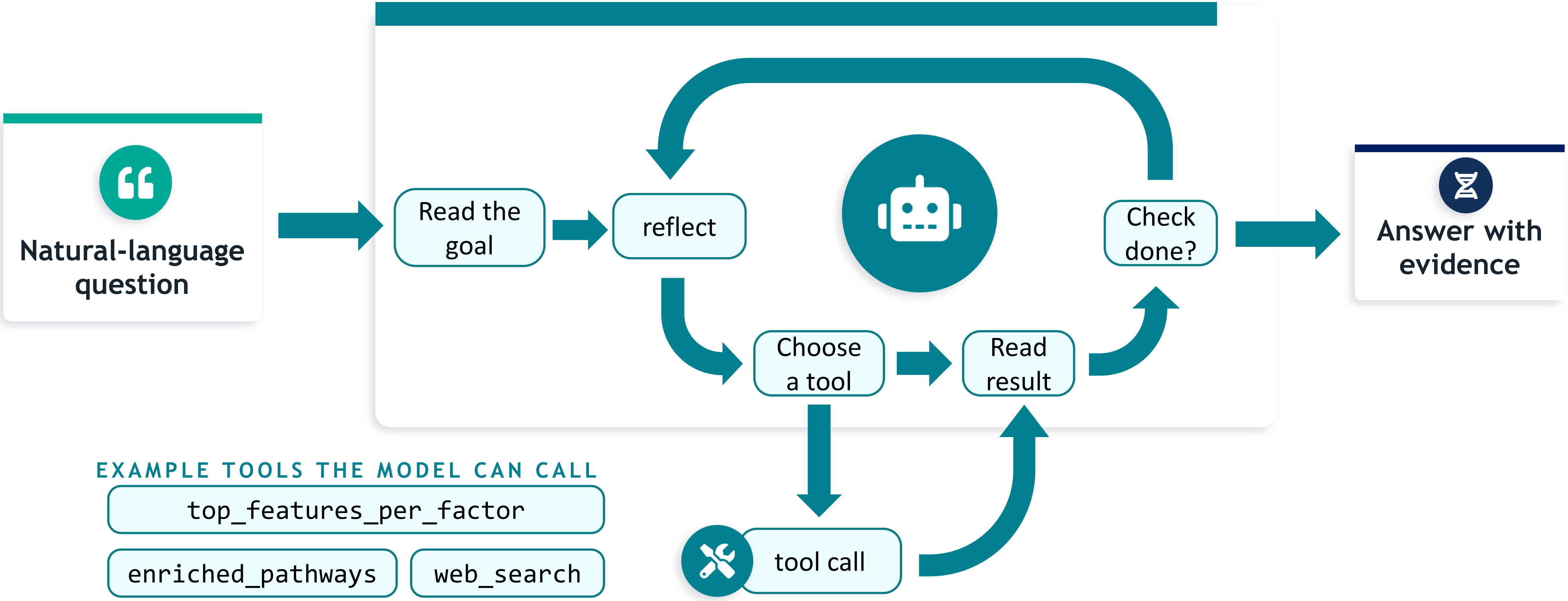

EXAMPLE TOOLS THE MODEL CAN CALL

top_features_per_factor

enriched_pathways

web_search

get_relevant_pubmed_articles

THE AGENT LOOP

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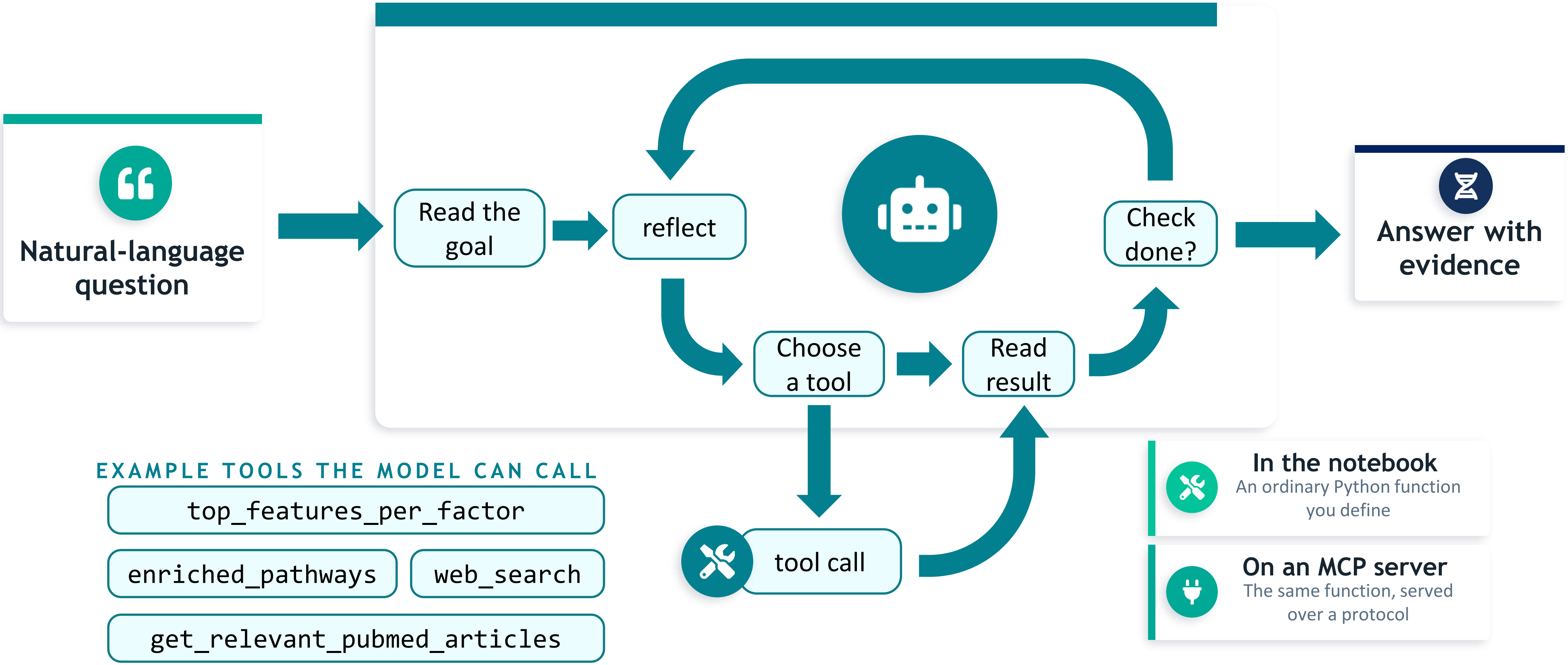

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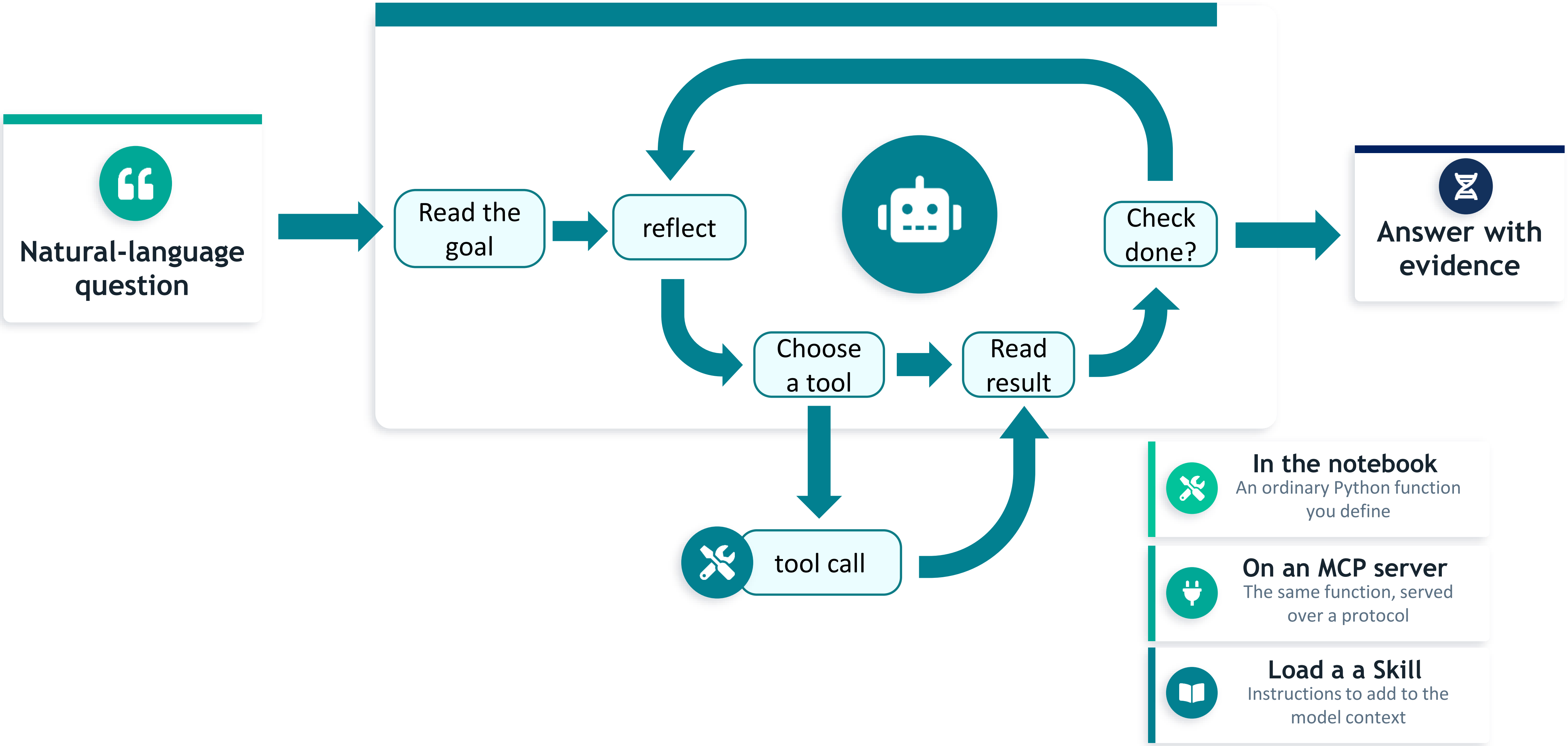

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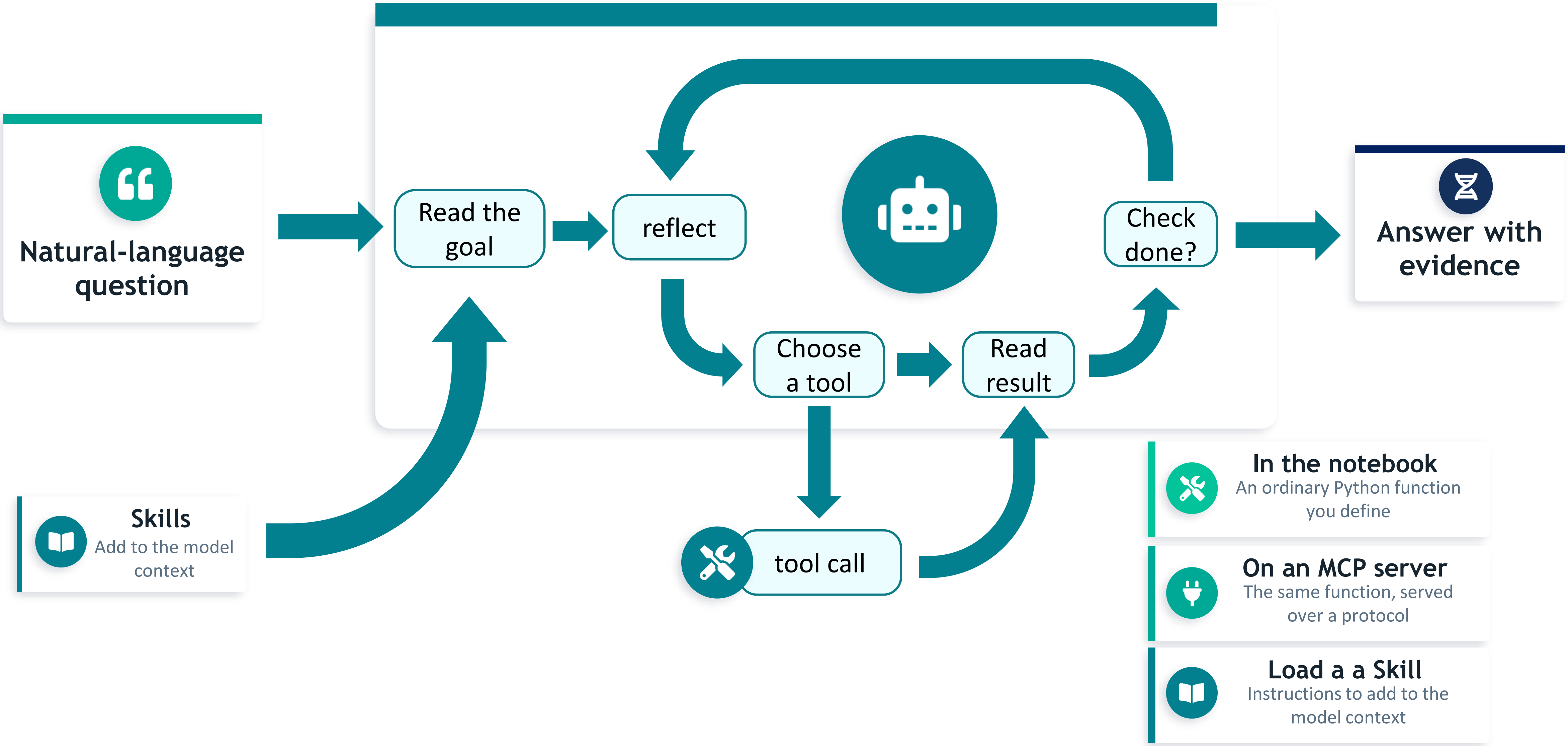

PRACTICAL

What you will do next

INTRODUCTION

Building Agentic LLM workflows

TRACK 1

Tools

Functions the model is allowed to call

TRACK 2

MCP

The same tools behind a portable server

TRACK 3

Skills

A written procedure the agent follows

SIDE BY SIDE

Comparison at a glance

Attribute	Tools	MCP	Skills
What it is	Plain functions the model calls	A protocol serving tools, data & prompts	Reusable behaviour instructions
It provides	Actions	Actions + resources + prompts, standardized	Procedural workflow
Setup cost	Low	Medium – High	Low
Portability	Notebook-bound	High — any compatible client	High — any agent
Gives real numbers?	Yes — via code	Yes — via served tools	No — directs tool use
Best for	Prototyping & teaching	Reuse across clients	Consistent, safe answers